

PRAMĀNA

Provenance-Ratified Attribution & Mitigation for Airsheds across Nations

A federated climate-action platform that does not just forecast pollution — it proves where it came from, in a form another country can independently verify.

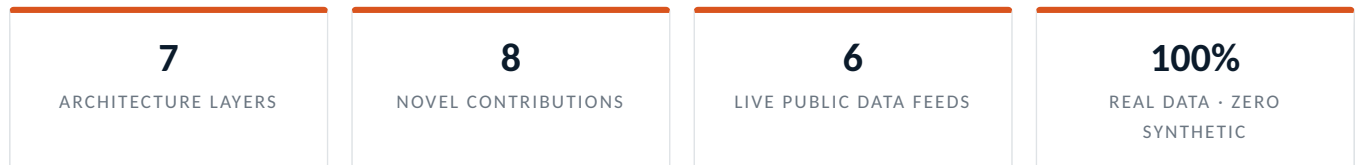
The thesis in one sentence: every BRICS nation already knows *how bad* its air is. None of them can prove, in real time and to a neighbour's satisfaction, *whose* it is — and that missing proof, not missing data, is why coordinated cross-border action never happens.

SECTION 01

Executive Summary

PRAMĀNA is an AI-powered federated climate action platform that fuses citizen-sourced photographs, national satellite imagery and meteorological data to detect hidden pollution hotspots, forecast air-quality spikes across economic corridors, and alert authorities — and then does the thing no existing system does: it issues a **cryptographically sealed, independently reproducible Attribution Certificate** naming the upwind sources responsible, with confidence intervals.

The problem statement asks for interoperability "so BRICS nations can share predictive models and coordinate resources." Every team in this track will build a dashboard that displays pollution. The winning insight is that **displaying pollution has never been the blocker**. Coordination fails because attribution across a political boundary is contested, unverifiable, and arrives months late in a journal paper. PRAMĀNA converts attribution from an academic post-mortem into an operational, auditable, real-time service — and it engineers the trust layer that makes a neighbouring state or nation willing to act on someone else's model output.



WHAT A JUDGE SEES IN NINETY SECONDS

A map of the Indo-Gangetic Plain. Fires ignite across Punjab. A plume advects south-east. Delhi's forecast crosses the GRAP Stage III threshold **42 hours before it happens**. The system does not merely warn — it opens a panel reading: "61.4% ± 7.2 of the projected PM_{2.5} excess is attributable to 3 named tehsils in south-east Punjab. Certificate `a3f9...c21`. Verify independently." The operator drags a slider to suppress those fires; the projected peak falls from 428 to 351 and the Stage III invocation disappears. Then the same pipeline is re-pointed at São Paulo and runs unchanged.

THE DIFFERENTIATING SENTENCE FOR THE PITCH

"Other systems tell a minister the air is bad. PRAMĀNA hands a minister a document that will survive a hostile cross-examination from the state next door — and that the state next door can re-run for themselves and get the identical answer."

SECTION 02

The Gap: What Already Exists and Why It Is Not Enough

Building novelty requires knowing precisely what is already deployed. Four mature capability clusters exist. None closes the loop.

Existing capability	What it does well	Where it stops
CPCB CAAQMS network + national AQI	Dense, official, hourly ground truth across Indian cities; open via the data.gov.in AQI API and aggregated globally by OpenAQ.	Measures concentration at a point. Says nothing about origin. Coverage collapses outside major cities — rural burning districts are effectively unmonitored.
Google Maps Platform Air Quality API	500 m resolution over 100+ countries, 70+ local indices, 30-day hourly history and 96-hour hourly forecast, heatmap tiles.	A superb <i>concentration</i> layer with no <i>causation</i> layer. It will tell you Delhi hits 420. It will not tell you which district's fires put it there, or what to shut down.
Academic source–receptor studies (HYSPLIT / CWT / CAMS)	Rigorous attribution. Published work has linked Punjab–Haryana fire counts to Delhi aerosol loading with $r \approx 0.78$ over October–November, and located specific high-contribution hotspots in south-east Punjab.	Retrospective, months-to-years after the episode, non-operational, non-reproducible by third parties, and openly contested — which is exactly why they change no policy.
GRAP / CAQM enforcement	A real statutory lever with four AQI-banded stages, now invoked pre-emptively on meteorological forecasts up to three days ahead.	Its actions are geographically blunt — blanket NCR-wide restrictions — because no operational system tells it <i>where</i> the pollution originated. It punishes the receptor, not the source.

The four gaps PRAMĀNA is built to close

1. **Attribution is offline.** Source apportionment exists in papers, not in APIs. There is no endpoint anywhere that answers "what fraction of tomorrow's Delhi peak is coming from outside Delhi, and from precisely where."
2. **Attribution is untrusted.** When one jurisdiction attributes pollution to another, the finding is immediately disputed — official fire-count reductions have been announced while receptor-city exceedance days kept rising. Without independent verifiability, attribution is just an accusation.
3. **Sovereignty blocks sharing.** "Share predictive models across BRICS nations" fails at first contact with the question "*why would we hand you our raw monitoring data?*" Any design that requires centralised raw data is dead on arrival at a ministry.
4. **Monitoring is blind at the source.** Reference-grade monitors sit in cities. The emissions are in rural districts with zero stations — the precise cells attribution most needs to resolve.

SECTION 03

The Novelty Stack — Eight Contributions

Each item below is scored for how much it separates us from a competent competing team. N1, N2 and N3 are the core; everything else amplifies them.

Airshed Source–Receptor Attribution Engine (SRAE)

An operational, hourly, multi-receptor attribution engine, not a retrospective study. For each receptor city we launch an ensemble of back-trajectories (3 arrival heights × 6 launch hours × 72 h lookback) over gridded wind fields, accumulate residence time per 0.1° upwind source cell to build a Concentration-Weighted Trajectory field, then decompose that field against three emission proxies:

- **Biomass** — VIIRS/MODIS Fire Radiative Power aggregated per cell from NASA FIRMS (near-real-time, ~3 h latency).
- **Industrial** — Sentinel-5P TROPOMI SO₂ and NO₂ column anomaly against a rolling 30-day cell baseline.
- **Urban / vehicular** — night-lights and road-density weighted NO₂ residual.

Output per receptor, per pollutant, per episode: a normalised **contribution vector** with bootstrap confidence intervals, aggregated upward from cell → tehsil → district → state → nation.

WHY IT IS NOVEL: the literature runs CWT retrospectively, on one receptor, one pollutant, one season. Nobody runs it live, multi-receptor, multi-source-class, decomposed by emission type, and serves it as a public API.

The Attribution Certificate and the Provenance Ledger

Every attribution PRAMĀNA emits is a signed, self-describing JSON artefact containing the contribution vector *and* a complete provenance block:

- SHA-256 of every input granule — TROPOMI asset IDs and acquisition timestamps, the exact FIRMS query and its response hash, the wind-field slice hash, station CSV hashes.
- Model version, code commit SHA, full parameter set, RNG seed, container digest.
- The computed vector, its confidence intervals, and the method identifier.

Certificates are chained into a Merkle tree; the daily root is published. Any party — a rival state pollution board, a neighbouring nation's ministry, an NGO, a journalist — runs `pramana verify <certificate>` and either reproduces the result bit-for-bit or receives a structured diff showing exactly which input diverged.

WHY IT WINS: this reframes the entire problem. Cross-border pollution attribution is not a data problem, it is a *trust* problem — and trust is an engineering deliverable. No competing team will make that argument, and it is the single most defensible thing you can say to a jury of policymakers, because every one of them has watched an attribution claim get dismissed as politically motivated.

N3 · CORE

Sovereignty-Preserving Federation

Each nation operates its own PRAMĀNA node. Raw monitoring data **never crosses a border**. Exactly three artefact classes cross:

- Model weight deltas for the shared spike forecaster, combined by federated averaging on Vertex AI.
- Attribution certificates (results plus provenance, never raw observations).
- Coarse gridded aggregates at a nationally-configurable resolution floor.

A node-level policy file declares what that nation is willing to emit. The federation is opt-in per artefact class.

WHY IT IS NOVEL: it is the only architecture in this track that survives the question a real ministry will ask. "Federated" in most submissions means "we drew a diagram." Ours means a demonstrated two-node federated averaging round between an India node and a second-nation node.

N4 · DEMO GOLD

Counterfactual Intervention Simulator

Because SRAE is a forward-composable model, source cells can be zeroed and the forecast re-run. The operator selects "suppress burning in these three tehsils for 48 hours" or "halt these four industrial clusters" and receives: projected peak AQI delta, GRAP stage avoided or retained, and **population exposure-hours averted** computed against gridded population.

WHY IT WINS: it converts a monitoring tool into a decision tool, and it is the most cinematic thirty seconds of the demo video. A slider that visibly drags a city out of "Severe" is worth more than three additional charts.

N5 · COVERAGE

Gemini-Powered Citizen Densification with Adversarial Trust Scoring

Rural source districts have no monitors. Citizens have cameras. The PWA captures a sky photograph; Gemini multimodal returns structured JSON — sky-visible fraction, haze severity band, visible plume or active burning, horizon-visibility class, self-reported confidence. In parallel the client computes classical dark-channel-prior transmission, local contrast, saturation and colour-attenuation features, which the published literature establishes as reliable predictors of $PM_{2.5}$. A regression head calibrated against the nearest reference station converts the fused features into a concentration estimate with an honest error bar.

The adversarial layer: because these readings will drive enforcement, they will be gamed. Each contributor carries a trust score derived from historical agreement with reference stations, EXIF and geolocation plausibility, perceptual-hash duplicate detection, and time-of-day/solar-geometry consistency. Low-trust readings are down-weighted, never silently dropped, and the weighting is recorded in the certificate.

WHY IT IS NOVEL: every citizen-science submission collects photos. Almost none reason about the fact that a crowd-sourced input feeding a punitive policy loop creates an immediate incentive to lie. Building the adversarial model into the trust layer signals genuine systems maturity.

N6 · CREDIBILITY

Benchmarked Against Google's Own Forecast — Publicly

PRAMĀNA's spike forecaster is evaluated against three baselines on identical historical windows: persistence, seasonal climatology, and the **Google Maps Platform Air Quality API 96-hour forecast**. The hypothesis under test is narrow and falsifiable: on *episodic transboundary fire-driven events*, explicitly ingesting upwind FRP and trajectory-weighted emission flux improves onset lead time and reduces peak error relative to a general-purpose forecaster that does not decompose upwind sources.

WHY IT WINS: benchmarking against the sponsor's own product and committing to report the number in either direction is the strongest possible credibility signal. It is also honest — and a jury that includes technical reviewers will notice immediately that we are the only team with a control condition.

N7 · ACTIONABILITY

The Pre-GRAP Advisory Object

The system's terminal output is not an AQI number. It is a structured advisory: predicted stage crossing and onset time with confidence, the attribution vector, a ranked list of *targeted* interventions with their simulated benefit, and an auto-drafted multilingual public notice — generated by Gemini, translated by the Translation API, and rendered to audio by Text-to-Speech for IVR delivery to farming districts in Punjabi and Hindi.

WHY IT MATTERS: it directly satisfies the problem statement's "alert relevant authorities for rapid intervention," and the multilingual voice path satisfies the build requirement for voice support where the track calls for it.

N8 · INTEROPERABILITY

The BRICS Airshed Interop Schema (open standard, not a claim)

We publish a versioned JSON Schema for observations, contribution vectors and certificates; an OpenAPI 3.1 specification for the node API; a permissive licence; and a written mapping against the nine Digital Public Goods Standard indicators. A conformance test suite lets any nation validate their own implementation.

WHY IT MATTERS: "designed for interoperability" and "digital public good" appear verbatim in the problem statement. Most teams will assert both on a slide. We ship the artefacts that make them true, which is a direct hit on the Cross-Border Applicability criterion (20%).

SECTION 04

System Architecture — Seven Layers

Layer	Name	Function	Google stack
L1	Ingest	Scheduled pulls of satellite, fire, meteorological, ground-station and baseline-forecast feeds; retry, backfill and granule-level hashing on the way in.	Cloud Functions + Cloud Scheduler; Earth Engine; Maps Platform Air Quality API
L2	Harmonise	Reprojection to a common 0.1° grid; TROPOMI quality-assurance filtering; unit normalisation; temporal alignment; a single canonical observation table.	BigQuery (partitioned + clustered), Earth Engine exports
L3	Sense	Citizen photo intake, Gemini structured extraction, dark-channel feature computation, station calibration, contributor trust scoring.	Gemini multimodal (Vertex AI), Firebase Auth + Storage + Firestore
L4	Forecast	72-hour spatiotemporal spike forecaster. Features: TROPOMI columns, trajectory-weighted upwind FRP, boundary-layer height, wind vectors, station lags, densified citizen field, Maps AQ baseline as a covariate.	Vertex AI custom training + model serving
L5	Attribute	SRAE. Trajectory ensemble → residence-time field → CWT → emission-proxy decomposition → contribution vector with bootstrap intervals.	Cloud Run jobs (vectorised NumPy), BigQuery
L6	Prove	Certificate assembly, input hashing, Merkle chaining, root publication, verifier CLI and public verification endpoint.	Cloud Run, BigQuery ledger table, Cloud Storage
L7	Act	Pre-GRAP advisory generation, counterfactual simulator, multilingual notice drafting and voice rendering, operator console, public map.	Gemini, Translation API, Text-to-Speech, Maps Platform, Cloud Run, Firebase Hosting

DESIGN RULE

Every layer writes its outputs with the hash of its inputs attached. Provenance is not a feature bolted on at L6 — it is a property of the data model from L1 onward. This is what makes bit-identical third-party reproduction achievable rather than aspirational.

SECTION 05

Data Sources — All Real, All Public

A deliberate constraint: **no synthetic data anywhere in the pipeline**. The build requirements ask for real or realistic data, and in this track — unlike the health or governance tracks — genuinely real data exists end to end. This is a structural advantage we should state explicitly in the pitch.

Source	Provides	Access	Role
Sentinel-5P TROPOMI (via Earth Engine)	NO ₂ , SO ₂ , CO, HCHO, aerosol index at 0.01°, 2018-07 to present, NRTI and OFFL	Earth Engine catalog	Plume detection, industrial anomaly, transport signal
NASA FIRMS (VIIRS S-NPP / NOAA-20/21, MODIS)	Active fire detections with FRP, 375 m, ~3 h latency, free MAP_KEY, area and country endpoints	REST API	Biomass emission proxy — the core attribution driver
CPCB CAAQMS	Official hourly station-level pollutant readings across India	data.gov.in AQI API	Ground truth, calibration, validation
OpenAQ	Globally harmonised aggregation of official networks including CPCB	REST API	Cross-border ground truth for non-India nodes
Google Maps Platform Air Quality API	500 m current / 30-day history / 96-hour forecast, 70+ indices, heatmap tiles, 100+ countries	GCP	Baseline forecaster to benchmark against; heatmap base layer; instant global coverage
MODIS AOD, NDVI (Earth Engine)	Column aerosol loading; harvest-phase proxy	Earth Engine catalog	Independent aerosol validation; burning-season context
Reanalysis wind / boundary-layer fields	u/v winds at multiple levels, PBL height, temperature	Earth Engine catalog (ERA5 family)	Trajectory advection and dispersion
Gridded population	Population per grid cell	Earth Engine catalog	Exposure-hours and health-burden weighting

Cross-border corridors selected for the demo. Primary: Punjab–Haryana → Delhi NCR (the world's best-documented transboundary smog corridor, giving us external literature to sanity-check against). Secondary, to prove portability without code changes: a second BRICS corridor driven by biomass burning — the pipeline is bounding-box parameterised, so the second corridor is a configuration file, not a fork.

SECTION 06

Google AI Integration — Mandatory Compliance Map

Rule 01 is absolute: submissions without Google AI integration are not considered, and criterion one of the rubric asks whether Google AI is doing *meaningful* work. Our answer spans generative, predictive and vision categories simultaneously.

Google product	Category	Doing what, specifically
Gemini (multimodal)	Vision + GenAI	Structured extraction from citizen sky photographs — haze band, plume presence, active-burning detection, sky fraction — emitted as schema-validated JSON at temperature 0
Vertex AI — custom training & serving	Predictive	The 72-hour spike forecaster; federated averaging across nation nodes; model registry providing the version pin recorded in every certificate
Google Earth Engine	Geospatial	TROPOMI, MODIS AOD/NDVI, reanalysis wind fields, population — server-side reduction and export to BigQuery
Google Maps Platform — Air Quality API	Geospatial	Benchmark baseline forecast; heatmap tile overlay; instant coverage for corridors outside our dense-data region
Google Maps Platform — Maps JS	Geospatial	Operator console, animated airshed flow map, replay theatre
BigQuery	Data	Canonical observation store, trajectory residence-time tables, the Merkle provenance ledger
Cloud Run / Cloud Functions	Backend	Ingest schedulers, SRAE compute jobs, node API, verifier service
Firebase	Backend	Auth, citizen photo storage, realtime alert fan-out, hosting
Cloud Translation API	Language	Advisories localised across BRICS linguistic regions
Cloud Text-to-Speech	Voice	IVR-ready audio advisories for low-literacy farming districts
Cloud Speech-to-Text	Voice	Voice-note citizen reports where typing is a barrier

SECTION 07

Validation Protocol — The Numbers That Win the Room

This is the section most teams will not have, and it is where the AI/Technical Execution weighting (25%) is actually decided. Nothing here is optional.

Historical replay harness

Replay complete post-monsoon episodes (late October to mid November) across multiple years. For each episode the harness reconstructs the exact information state available at each forecast issue time — no future leakage — and scores what PRAMĀNA would have said against what actually occurred.

Metric family	Specific measure	Compared against
Spike detection skill	Probability of detection, false alarm ratio, critical success index for each GRAP stage crossing	Persistence; climatology; Google AQ API forecast
Warning lead time	Median hours between first correct stage-crossing alert and actual crossing	Same three baselines
Peak accuracy	MAE and bias on episode peak AQI and peak PM _{2.5}	Same three baselines
Attribution plausibility	Correlation of computed upwind biomass contribution against independent fire-count and AOD series; agreement with published corridor findings	Peer-reviewed literature values
Reproducibility	Certificates re-verified bit-identically from a clean container by an independent run	Target: N of N, zero divergence
Citizen model	MAE of photo-derived PM _{2.5} versus nearest reference station, stratified by trust tier	Station reading
Portability	Corridors and nations executed with zero code change	Target: ≥ 2 nations

NON-NEGOTIABLE DISCIPLINE

Report the numbers honestly, including where a baseline beats us. A team that says "we improved lead time on fire-driven events by X hours but did not beat the Google baseline on ordinary winter stagnation, and here is why" is vastly more credible than one claiming uniform superiority. Judges evaluating deployability are specifically listening for whether you know your system's failure modes.

SECTION 08

Rubric Mapping — Where Each Point Comes From

Criterion	Weight	How PRAMĀNA scores maximally
AI / Technical Execution	25%	Three Google AI categories doing non-decorative work; a novel attribution algorithm; a validated forecaster benchmarked against a control; end-to-end functioning flow with a deployed link
Problem-Solution Fit	20%	Every clause of the challenge text is implemented: citizen photos, local readings, satellite imagery, meteorological data, hidden hotspot detection, corridor spike forecasting, authority alerting, model sharing, interoperability
Cross-Border Applicability	20%	Global data sources by construction; a live second-nation corridor demonstrated; federation that preserves sovereignty; a published interop schema and conformance suite
Deployability & Scalability	20%	Entirely managed GCP services; node-per-nation deployment model; the pilot unit is one airshed and can be stood up in weeks; certificates give the audit trail a ministry legally requires
Impact Potential	10%	Exposure-hours averted quantified against gridded population; targeted enforcement replaces blanket economic shutdowns — the benefit is measured in both health and avoided economic cost
Presentation & Clarity	5%	One sentence an international delegate understands instantly: <i>proof of who caused it, that the accused can verify themselves</i>

SECTION 09

Feature Catalogue and Build Priority

Feature	Priority	Note
L1-L2 ingest and harmonisation for primary corridor	MUST SHIP	Everything depends on it; build first, build boring
SRAE attribution engine (N1)	MUST SHIP	The core contribution — human-owned, not delegated
Attribution Certificate + Merkle ledger + verifier (N2)	MUST SHIP	The differentiator; without it we are another dashboard
72-hour spike forecaster + replay harness (N6)	MUST SHIP	Validation numbers are the credibility spine
Operator console: airshed flow map + attribution panel	MUST SHIP	The demo surface
Gemini citizen photo intake + calibration (N5)	CORE	Satisfies "citizen-sourced data" explicitly
Counterfactual simulator (N4)	CORE	Highest demo value per hour of work
Second-nation corridor	CORE	20% of the rubric hinges on demonstrating this live
Pre-GRAP advisory + multilingual notice + TTS (N7)	CORE	Closes the loop to "alert authorities"
Two-node federated averaging round (N3)	CORE	One real round beats ten diagram slides
Interop schema, OpenAPI spec, DPG mapping (N8)	CORE	Cheap to produce, disproportionate rubric payoff
Contributor trust scoring (N5 adversarial layer)	STRETCH	Ship at least duplicate-hash + geo-plausibility checks
Episode Replay Theatre (scrubbable historical playback)	STRETCH	Extraordinary for the demo video if time allows
Voice-note citizen reports via Speech-to-Text	STRETCH	Nice accessibility story; low rubric weight
Health burden overlay (exposure-hours by population)	STRETCH	Feeds the Impact criterion directly
Public certificate verification web page	STRETCH	Turns N2 from a claim into something a judge can click

Execution Plan and Guildly Agent Allocation

The build runs as parallel lanes coordinated through Guildly, with one hard rule stated up front.

THE ONE RULE THAT PROTECTS THE WIN

L5 (SRAE) and L6 (certificates) stay human-authored. These are the novelty. You will be cross-examined on them at Demo Day and you must be able to defend every design decision without hesitation. Delegate ingestion, frontend, glue code, documentation and test scaffolding freely — never delegate the thing that makes you different.

Guildly agent	Owns	Definition of done
Manager	Sprint board, critical path, blocker escalation, daily cut decisions	Every task tracked; scope cuts made early, not on the last night
Product Manager	PRD, deliverable checklist, DPG mapping, interop schema documentation	All five submission artefacts specified before build starts
SDE — Data	L1 ingest connectors, L2 harmonisation, BigQuery schema, backfill jobs	Canonical observation table populated for both corridors; hashes recorded
SDE — ML	L4 forecaster, feature pipeline, replay harness, baseline comparisons	Metrics table populated with real numbers against all three baselines
Human (you)	L5 SRAE, L6 certificate and ledger, verifier, federation round	Attribution reproducible bit-identically from a clean container
SDE — Frontend	Operator console, Maps flow map, counterfactual slider, citizen PWA	Deployed public link, working on mobile and desktop
Designer	Pitch deck, demo storyboard, diagram set, visual language	10–12 slides matching the demo narrative beat for beat
Reviewer	Reproducibility audit, third-party licence and citation compliance (Rule 03), originality check (Rule 02)	Every reused component licensed and cited in the repository

Day plan

DAY 1 — Foundation and de-risking

Register the team immediately. Stand up the GCP project, enable Earth Engine, Vertex AI, Maps Platform Air Quality, BigQuery, Cloud Run. Obtain the FIRMS MAP_KEY and the data.gov.in AQI key. Build L1 connectors and prove one end-to-end granule lands in BigQuery with its hash. Define the certificate JSON schema on paper before writing the engine. **Kill risk early:** if Earth Engine authentication or Maps AQ quota is going to bite, it must bite today, not on day four.

DAY 2 — The engine

Harmonisation to the common grid with TROPOMI quality filtering. Trajectory ensemble implementation, vectorised. Residence-time and CWT fields. First raw contribution vector for one receptor on one historical episode — even if crude, get a number on the screen. Forecaster feature pipeline built in parallel by the ML lane.

DAY 3 — Proof and prediction

Certificate builder, hashing, Merkle chain, verifier CLI. Forecaster trained; replay harness executing; first metrics against persistence and the Google AQ API baseline. Frontend lane brings up the map and attribution panel against live API responses. Second corridor configured and ingesting.

DAY 4 — Amplifiers

Counterfactual simulator. Gemini citizen photo path with structured output and station calibration. Two-node federated averaging round executed and logged. Pre-GRAP advisory generation with Translation and TTS. Publish interop schema and OpenAPI spec. Freeze features at end of day — anything not working by now is cut, not rescued.

DAY 5 — Ship

Final validation run with all metrics locked and written into the README. Deploy to a stable public URL and smoke-test it from a device you have never used. Record the 3–5 minute demo video against the beat sheet below. Finish the 10–12 slide deck. Write the 2–3 line description. Verify the repository is public or access-granted, licensed, and every third-party component cited. Submit with hours to spare, not minutes.

SECTION 11

The Demo Video — Beat Sheet

The video is scored under Presentation & Clarity and is the artefact most likely to be watched in full by every judge. Structure it as a narrative, not a feature tour.

- 0:00–0:25** **The stake, not the statistic.** Post-monsoon Delhi. GRAP invoked. Construction halted, trucks banned, schools shut — across the receptor city, while the sources sit untouched a hundred kilometres upwind. State the problem as a governance failure, not an air-quality failure.
- 0:25–0:50** **The gap.** Show a live Google Air Quality API forecast for Delhi. It is accurate and it is useless for enforcement, because it contains no causal information. Name the gap explicitly.
- 0:50–1:40** **PRAMĀNA detects.** Fires ignite on the FIRMS layer. TROPOMI plume advects. The forecaster crosses the Stage III threshold with the lead time displayed against the persistence and Google baselines side by side. Real numbers on screen.
- 1:40–2:30** **PRAMĀNA attributes.** The airshed flow map animates transport from source cells to receptor. The contribution vector resolves with confidence intervals, drilling from nation to state to named tehsil.
- 2:30–3:10** **The proof — the moment that wins.** Open the certificate. Show the input hashes and model pin. Run the verifier in a clean container on screen and reproduce the result bit-identically. Say the line: *"the state being named can check this themselves, and get the same answer."*
- 3:10–3:50** **PRAMĀNA acts.** Counterfactual slider suppresses the three named tehsils; the projected peak drops out of Severe. The advisory generates in Hindi and Punjabi and plays as audio.
- 3:50–4:30** **It crosses borders.** Change the corridor configuration. The identical pipeline runs over a second BRICS nation with no code change. Show the federated round: weights moved, raw data did not.
- 4:30–5:00** **Close on deployability.** Managed services, node per nation, one airshed as the pilot unit, open schema, published metrics including where we lose. End on the certificate, not the dashboard.

Draft two-line description

PRAMĀNA is a federated climate-action platform that fuses citizen photographs, satellite and meteorological data to forecast air-quality spikes across BRICS economic corridors — and then issues a cryptographically verifiable Attribution Certificate identifying which upwind sources caused them. Because any nation can independently reproduce any attribution bit-for-bit, it turns cross-border pollution from a contested accusation into shared, auditable evidence that authorities can act on.

Deck outline (10–12 slides)

1. The governance failure, stated in one image
2. What exists today and the four gaps
3. PRAMĀNA in one sentence
4. Architecture — the seven layers
5. N1: how attribution actually works
6. N2: the certificate and why trust is the real blocker
7. N3: federation without surrendering sovereignty
8. Validation — the metrics table, including the honest losses
9. From forecast to enforcement: the Pre-GRAP advisory and counterfactual
10. Cross-border: the second corridor and the interop schema
11. Deployment path — pilot unit, cost, timeline in weeks
12. Impact — exposure-hours averted, economic cost of blanket shutdowns avoided

SECTION 12

Risks and Mitigations

Risk	Severity	Mitigation
Earth Engine auth or export latency stalls the pipeline	High	Prove the path on Day 1; pre-export episode windows to BigQuery rather than querying Earth Engine live during the demo
Maps Air Quality API quota or billing limits	Medium	Cache all baseline forecasts to BigQuery on first fetch; the demo reads cache, never live quota
Trajectory computation too slow for interactive use	High	Precompute residence-time fields per episode offline; the console reads precomputed tensors; vectorise rather than loop
Gemini output variance breaks downstream parsing	Medium	Structured JSON schema, temperature 0, strict validation with a typed fallback path; never let a malformed response reach the forecaster
Attribution reads as politically inflammatory	High	Language discipline throughout: "contribution register", never "blame"; always publish confidence intervals; frame output as evidence for joint action, never as an indictment. Say this explicitly in the pitch — it shows political literacy
Scope overrun sinks the submission	High	The must-ship / core / stretch split in Section 09 is binding. Feature freeze at end of Day 4 regardless of state
Rule 02 originality challenge	Medium	Fresh repository, commits dated within the hackathon window, and a written statement of what is new versus what is prior art with citations
Rule 03 licence compliance	Low	Reviewer agent maintains a third-party licence and attribution file continuously, not retroactively

SECTION 13

Beyond the Hackathon

Pilot path. The natural deployment unit is a single airshed with one receptor city and its upwind source districts. That is small enough for a state pollution control board to adopt without national coordination, and the certificate mechanism is precisely what makes an inter-state agreement enforceable. The problem statement's own FAQ confirms that selected solutions are routed through official government channels for ministry-level pilot evaluation.

Research path. There is a genuine publishable contribution here that is independent of whether you win: *operational, third-party-verifiable source-receptor attribution for transboundary particulate episodes*. The atmospheric science community has the attribution methods and the computer science community has the provenance methods; almost nobody has joined them. Target atmospheric environment or environmental modelling venues rather than the aerospace track you are already working — this is a different community. Keep the replay harness and the certificate corpus clean from day one and the paper is largely written by the time the hackathon ends.

FINAL FRAMING

Every competing team in Track 02 will build something that watches the air. Build the thing that settles the argument about it. That is the difference between a dashboard and an institution.

Key Sources Consulted

1. Google Earth Engine Data Catalog — Sentinel-5P TROPOMI collections (NO₂, SO₂, CO, HCHO, aerosol index; 0.01°; 2018-07 onward).
developers.google.com/earth-engine/datasets/catalog/sentinel-5p
2. Google Maps Platform — Air Quality API overview and release notes (500 m resolution, 100+ countries, 70+ indices, 30-day history, 96-hour forecast, heatmap tiles). developers.google.com/maps/documentation/air-quality
3. NASA LANCE FIRMS — Area, Country and Data Availability API documentation; VIIRS 375 m and MODIS active fire products; free MAP_KEY registration.
firms.modaps.eosdis.nasa.gov/api
4. OpenAQ API documentation and platform (global harmonised aggregation including CPCB). docs.openaq.org
5. CPCB real-time AQI API via data.gov.in; CPCB CCR portal for station-level archives.
6. Commission for Air Quality Management — Graded Response Action Plan for NCR, staged AQI thresholds and forecast-based pre-emptive invocation.
caqm.nic.in
7. "Determining hotspots of gaseous criteria air pollutants in Delhi airshed and its association with stubble burning," *Scientific Reports* — TROPOMI-based back-trajectory hotspot identification in south-east Punjab.
8. "Tracing the haze: satellite-based assessment of stubble burning and air quality in Delhi," *Air Quality, Atmosphere & Health* — HYSPLIT trajectories, 36–48 h Punjab-to-Delhi transport, fire counts explaining ~78% of Delhi AOD variance in October–November.
9. "Does Stubble Burning Really Contribute in Delhi's Air Pollution?", *Water, Air, & Soil Pollution* — multi-source evidence combining ground, model and satellite data.
10. "Uncovering local aggregated air quality index with smartphone captured images," *Scientific Reports*; and HVAQ dataset (arXiv) — dark channel prior, transmission, contrast and saturation features for image-based PM_{2.5} estimation.
11. "Measuring PM_{2.5} Concentrations from a Single Smartphone Photograph," *Remote Sensing* — cue combination and extinction-index methodology.